JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2025 3rd April Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry	25 questions
Mathematics	25 questions
Physics	25 questions
Answer key	Included at the end

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QR codes open the DekhoExams website.

Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

The correct order of the complexes $[Co(NH_3)_5(H_2O)]^{3+}(A)$, $[Co(NH_3)_6]^{3+}(B)$, $[Co(CN)_6]^{3-}(C)$ and $[CoCl(NH_3)_5]^{2+}(D)$ in terms of wavelength of light absorbed is

(A) $D > C > B > A$

(B) $C > B > A > D$

(C) $D > A > B > C$

(D) $C > B > D > A$

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Q2. MCQ (Single Correct Answer)

Which of the following statements are correct?

A. The process of adding an electron to a neutral gaseous atom is always exothermic.

B. The process of removing an electron from an isolated gaseous atom is always endothermic.

C. The 1st ionization energy of boron is less than that of beryllium.

D. The electronegativity of C is 2.5 in CH_4 and CCl_4

E. Li is the most electropositive among elements of group I.

Choose the correct answer from the options given below:

(A) A, C and D Only

(B) B and C Only

(C) B and D Only

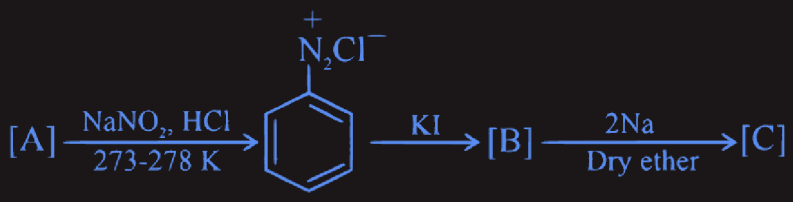
(D) B, C and E Only

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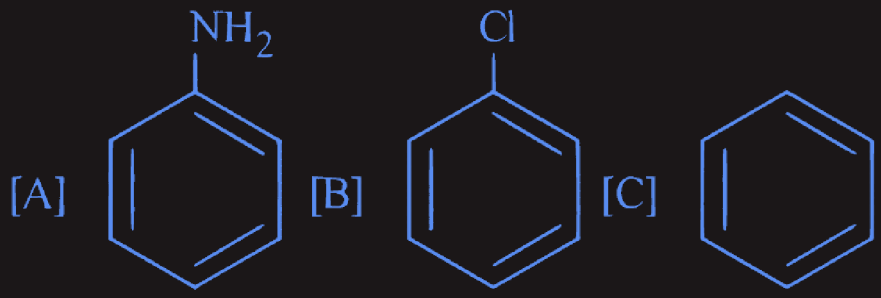
Q3. MCQ (Single Correct Answer)

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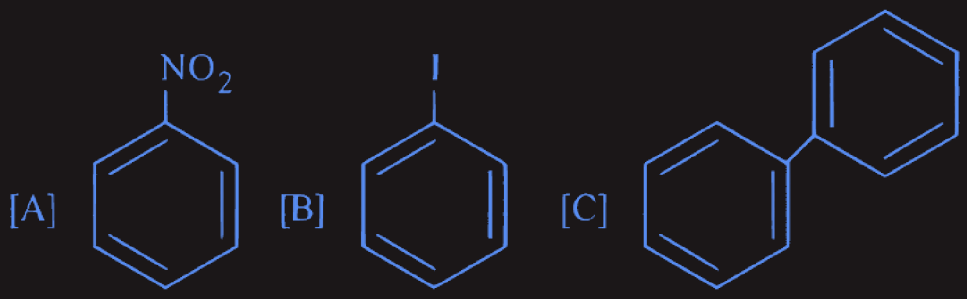
Identify [A], [B] and [C], respectively in the following reaction sequence :



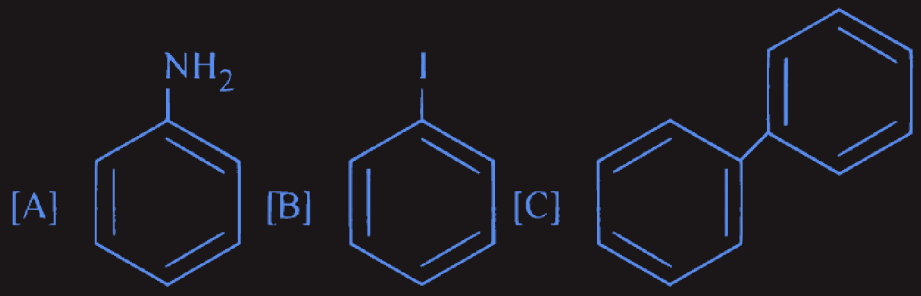
(A)



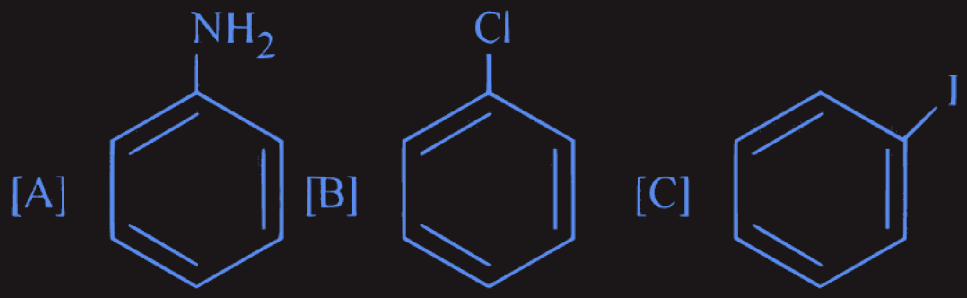
(B)



(C)



(D)



Q4. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

LIST-I (Molecules/ion)		LIST-II (Hybridisation of central atom)	
A.	PF_5	I	dsp^2
B	SF_6	II	$sp^3\,d$
C	$Ni(CO)_4$	III	$sp^3\,d^2$
D	$[PtCl_4]^{2-}$	IV	sp^3

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-III, B-I, C-IV, D-II

(C) A-II, B-III, C-IV, D-I

(D) A-I, B-II, C-III, D-IV

Q5. MCQ (Single Correct Answer)

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Which of the following postulate of Bohr's model of hydrogen atom is not in agreement with quantum mechanical model of an atom?

(A) The electron in a H atom's stationary state moves in a circle around the nucleus.

(B) An atom can take only certain distinct energies E_1, E_2, E_3, etc. These allowed states of constant energy are called the stationary states of atom.

(C) An atom in a stationary state does not emit electromagnetic radiation as long as it stays in the same state.

(D) When an electron makes a transition from a higher energy stationary state to a lower energy stationary state, then it emits a photon of light.

Q6. MCQ (Single Correct Answer)

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In the following system, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ at equilibrium, upon addition of xenon gas at constant T and p, the concentration of

(A) PCl_5, PCl_3 and Cl_2 remain constant

(B) PCl_3 will increase

(C) Cl_2 will decrease

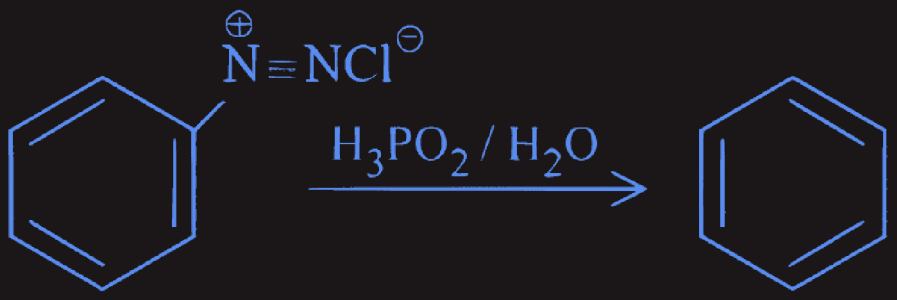
(D) PCl_5 will increase

Q7. MCQ (Single Correct Answer)

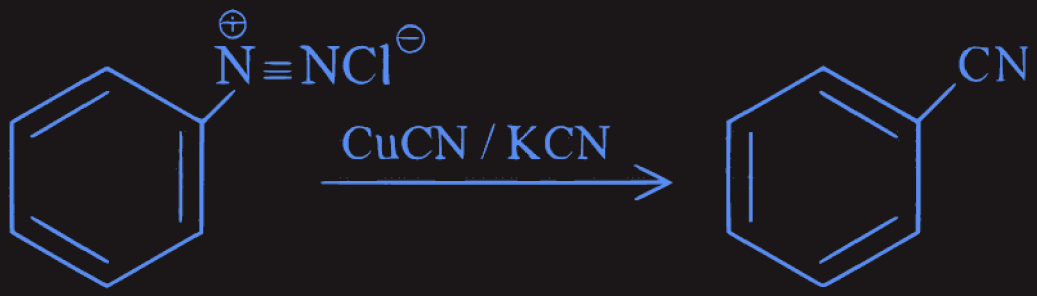
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In the following reactions, which one is NOT correct?

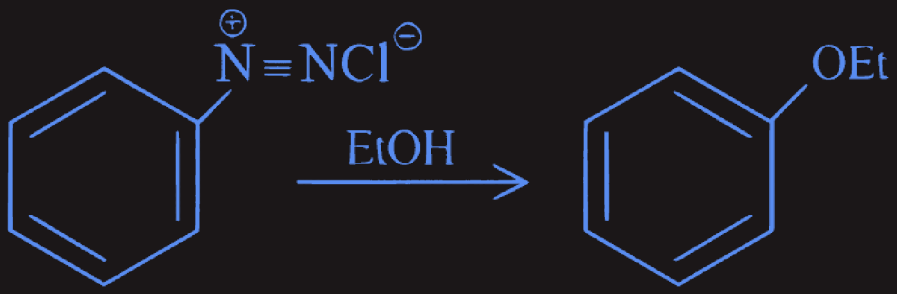
(A)



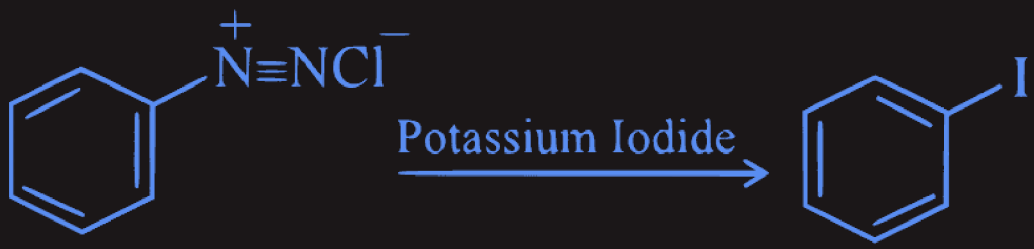
(B)



(C)



(D)



Q8. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : The N - N single bond is weaker and longer than that of P - P single bond.

Statement II : Compounds of group 15 elements in +3 oxidation states readily undergo disproportionation reactions.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

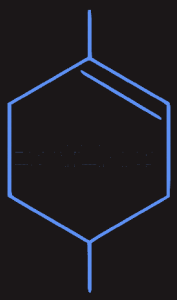
(D) Statement I is true but Statement II is false

Q9. MCQ (Single Correct Answer)

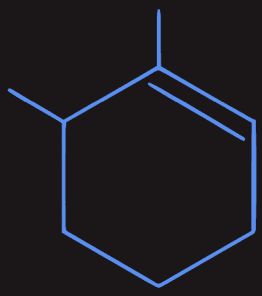
Which compound would give 3-methyl-6-oxoheptanal upon ozonolysis?

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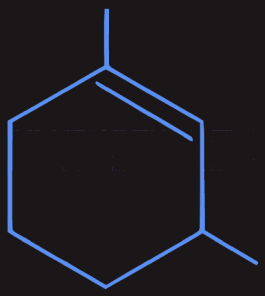
(A)



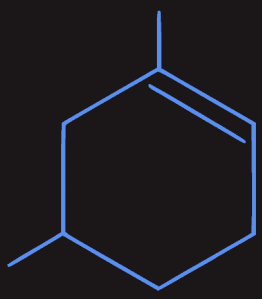
(B)



(C)



(D)



Q10. MCQ (Single Correct Answer)

2 moles each of ethylene glycol and glucose are dissolved in 500 g of water. The boiling point of the resulting solution is: (Given : Ebullioscopic constant of water $= 0.52\ \mathrm{K\ kg\ mol^{-1}}$)

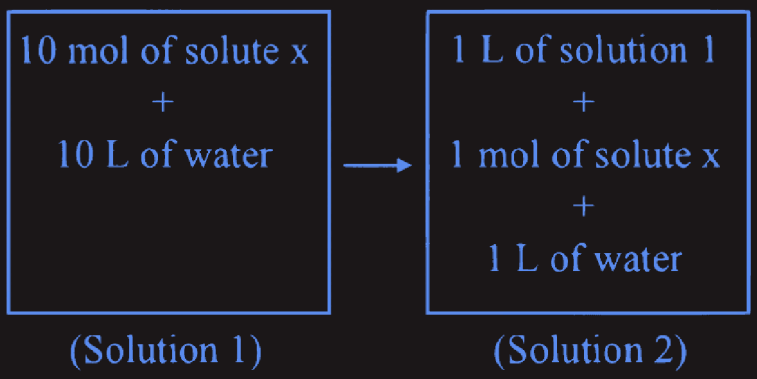
(A) 377.3 K (B) 379.2 K (C) 375.3 K (D) 277.3 K

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Q11. MCQ (Single Correct Answer)

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Which of the following properties will change when system containing solution 1 will become solution 2 ?



(A) Molar heat capacity

(B) Concentration

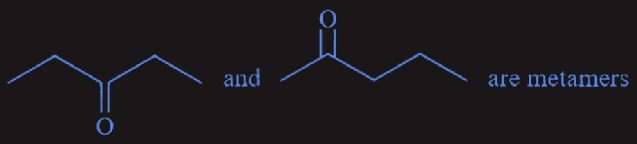
(C) Gibbs free energy

(D) Density

Q12. MCQ (Single Correct Answer)

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Identify the correct statements from the following.

A.  are metamers

B.  are functional isomers

C.  are position isomers

D.  are homologous

Choose the correct answer from the options given below:

(A) B & C Only

(B) C & D Only

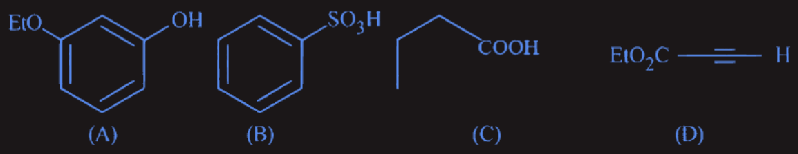
(C) A,B & C Only

(D) A & B Only

Q13. MCQ (Single Correct Answer)

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The least acidic compound, among the following is:



(A) B

(B) D

(C) A

(D) C

Q14. MCQ (Single Correct Answer)

Among 10^{-9} g (each) of the following elements, which one will have the highest number of atoms?

Element: $\mathrm{Pb}, \mathrm{Po}, \mathrm{Pr}$ and Pt

(A) Pr (B) Pt (C) Po (D) Pb

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Q15. MCQ (Single Correct Answer)

The metal ions that have the calculated spin-only magnetic moment value of 4.9 B.M. are :

A. Cr^{2+}

B. Fe^{2+}

C. Fe^{3+}

D. Co^{2+}

E. Mn^{3+}

Choose the correct answer from the options given below:

(A) A, D and E Only (B) A, B and E Only (C) A, C and E Only (D) B and E Only

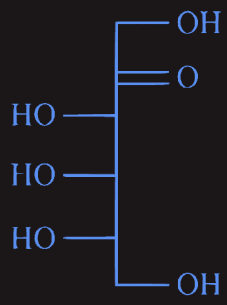
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Q16. MCQ (Single Correct Answer)

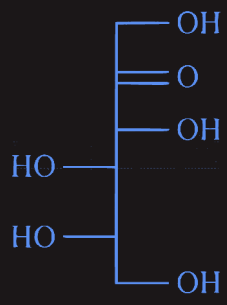
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Which of the following is the correct structure of L-Fructose?

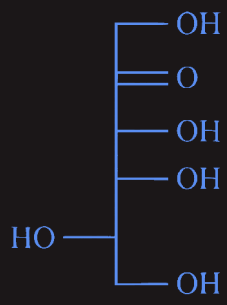
(A)



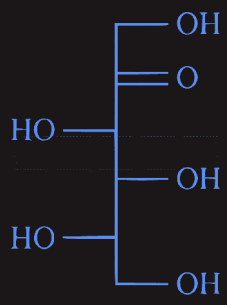
(B)



(C)



(D)



Q17. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : A catalyst cannot alter the equilibrium constant (K_c) of the reaction, temperature remaining constant.

Statement II : A homogeneous catalyst can change the equilibrium composition of a system, temperature remaining constant.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q18. MCQ (Single Correct Answer)

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Correct order of limiting molar conductivity for cations in water at 298 K is :

(A) $H^+ > Na^+ > K^+ > Ca^{2+} > Mg^{2+}$

(B) $H^+ > Ca^{2+} > Mg^{2+} > K^+ > Na^+$

(C) $Mg^{2+} > H^+ > Ca^{2+} > K^+ > Na^+$

(D) $H^+ > Na^+ > Ca^{2+} > Mg^{2+} > K^+$

Q19. MCQ (Single Correct Answer)

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In a reaction $A + B \rightarrow C$, initial concentrations of A and B are related as $[A]_0 = 8[B]_0$. The half lives of A and B are 10 min and 40 min , respectively. If they start to disappear at the same time, both following first order kinetics, after how much time will the concentration of both the reactants be same?

(A) 20 min

(B) 40 min

(C) 80 min

(D) 60 min

Q20. MCQ (Single Correct Answer)

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Number of molecules from below which cannot give iodoform reaction is :

Ethanol, Isopropyl alcohol, Bromoacetone, 2-Butanol, 2-Butanone, Butanal, 2-Pentanone, 3-Pentanone, Pentanal and 3-Pentanol.

(A) 4

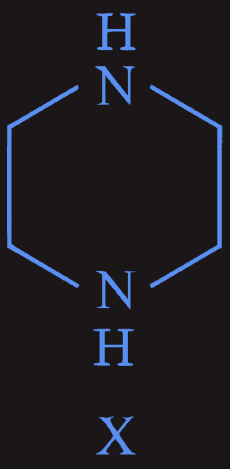
(B) 3

(C) 5

(D) 2

Q21. Numerical

During estimation of nitrogen by Dumas' method of compound $\mathrm{X}(0.42\ \mathrm{g})$



_____mL of $\mathrm{N_2}$ gas will be liberated at STP. (nearest integer)

(Given molar mass in $\mathrm{g\ mol^{-1}}$: $\mathrm{C}: 12, \mathrm{H}: 1,\ \mathrm{N}: 14$)

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Q22. Numerical

Consider the following reactions

$\mathrm{A} + \underset{\substack{\text{Little}\\\text{amount}}}{\mathrm{NaCl}} + \mathrm{H_2SO_4} \rightarrow \mathrm{CrO_2Cl_2} +$ Side Products

$\mathrm{CrO_2Cl_2}_{\text{ (Vapour)}} + \mathrm{NaOH} \rightarrow \mathrm{B} + \mathrm{NaCl} + \mathrm{H_2O}$

$\mathrm{B} + \mathrm{H^+} \rightarrow \mathrm{C} + \mathrm{H_2O}$

The number of terminal ' O ' present in the compound ' C ' is_____

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Q23. Numerical

0.5 g of an organic compound on combustion gave 1.46 g of $\mathrm{CO_2}$ and 0.9 g of $\mathrm{H_2O}$. The percentage of carbon in the compound is _____. (Nearest integer)

[Given : Molar mass (in $\mathrm{gmol^{-1}}$) $\mathrm{C}: 12, \mathrm{H}: 1, \mathrm{O}: 16$]

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Q24. Numerical

The number of optical isomers exhibited by the iron complex (A) obtained from the following reaction is_____.

$\mathrm{FeCl_3} + \mathrm{KOH} + \mathrm{H_2C_2O_4} \rightarrow \mathrm{A}$

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Q25. Numerical

Given :

$\Delta H^{\ominus}_{\text{sub}}[\text{C (graphite)}] = 710\ \text{kJ mol}^{-1}$

$\Delta_{C-H}H^{\ominus} = 414\ \text{kJ mol}^{-1}$

$\Delta_{H-H}H^{\ominus} = 436\ \text{kJ mol}^{-1}$

$\Delta_{C=C}H^{\ominus} = 611\ \text{kJ mol}^{-1}$

The $\Delta H_f^{\ominus}$ for $CH_2 = CH_2$ is_____ kJmol^{-1} (nearest integer value)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

If the domain of the function $f(x) = \log_e\left(\frac{2x-3}{5+4x}\right) + \sin^{-1}\left(\frac{4+3x}{2-x}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + 4\beta$ is equal to

(A) 4 (B) 3 (C) 7 (D) 5

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Q27. MCQ (Single Correct Answer)

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive numbers. If $a_3a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to

(A) 128 (B) 129 (C) 131 (D) 130

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Q28. MCQ (Single Correct Answer)

The sum $1 + 3 + 11 + 25 + 45 + 71 + \ldots$ upto 20 terms, is equal to

(A) 7240 (B) 8124 (C) 7130 (D) 6982

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Q29. MCQ (Single Correct Answer)

The number of solutions of the equation $2x + 3\tan x = \pi, x \in [-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$ is:

(A) 4 (B) 5 (C) 3 (D) 6

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Q30. MCQ (Single Correct Answer)

Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $0 \le x^2 + 2y \le 4$. Let l be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation. Then $l + m$ is equal to

(A) 18 (B) 20 (C) 17 (D) 19

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Q31. MCQ (Single Correct Answer)

The radius of the smallest circle which touches the parabolas $y = x^2 + 2$ and $x = y^2 + 2$ is

(A) $\frac{7\sqrt{2}}{16}$ (B) $\frac{7\sqrt{2}}{8}$ (C) $\frac{7\sqrt{2}}{2}$ (D) $\frac{7\sqrt{2}}{4}$

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Q32. MCQ (Single Correct Answer)

Let $f(x) = \int x^3\sqrt{3 - x^2}dx$. If $5f(\sqrt{2}) = -4$, then $f(1)$ is equal to

(A) $-\frac{6\sqrt{2}}{5}$ (B) $-\frac{8\sqrt{2}}{5}$ (C) $-\frac{2\sqrt{2}}{5}$ (D) $-\frac{4\sqrt{2}}{5}$

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Q33. MCQ (Single Correct Answer)

Let the domain of the function $f(x) = \log_2 \log_4 \log_6 \left(3 + 4x - x^2\right)$ be (a, b). If $\int_0^{b-a} \left[x^2\right] dx = p - \sqrt{q} - \sqrt{r}, p, q, r \in \mathbb{N}, \gcd(p, q, r) = 1$, where $[\cdot]$ is the greatest integer function, then $p + q + r$ is equal to

(A) 10 (B) 11 (C) 9 (D) 8

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Q34. MCQ (Single Correct Answer)

Let α and β be the roots of $x^2 + \sqrt{3}x - 16 = 0$, and γ and δ be the roots of $x^2 + 3x - 1 = 0$. If $P_n = \alpha^n + \beta^n$ and $Q_n = \gamma^n + \delta^n$, then $\frac{P_{25} + \sqrt{3}P_{24}}{2P_{23}} + \frac{Q_{25} - Q_{23}}{Q_{24}}$ is equal to

(A) 4 (B) 3 (C) 5 (D) 7

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Q35. MCQ (Single Correct Answer)

Let a line passing through the point $(4, 1, 0)$ intersect the line $\mathrm{L}_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at the point $A(\alpha, \beta, \gamma)$ and the line $\mathrm{L}_2 : x - 6 = y = -z + 4$ at the point $B(a, b, c)$. Then $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ is equal to

(A) 16 (B) 6 (C) 8 (D) 12

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Q36. MCQ (Single Correct Answer)

Let $f(x) = \begin{cases} (1 + ax)^{1/x} & , x < 0 \\ 1 + b, & x = 0 \\ \frac{(x+4)^{1/2} - 2}{(x+c)^{1/3} - 2}, & x > 0 \end{cases}$ be continuous at $x = 0$. Then $e^a bc$ is equal to:

(A) 64 (B) 48 (C) 36 (D) 72

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Q37. MCQ (Single Correct Answer)

A line passing through the point $P(\sqrt{5}, \sqrt{5})$ intersects the ellipse $\frac{x^2}{36} + \frac{y^2}{25} = 1$ at A and B such that $(PA) \cdot (PB)$ is maximum. Then $5\left(PA^2 + PB^2\right)$ is equal to :

(A) 290 (B) 377 (C) 338 (D) 218

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Q38. MCQ (Single Correct Answer)

Line L_1 passes through the point $(1, 2, 3)$ and is parallel to z-axis. Line L_2 passes through the point $(\lambda, 5, 6)$ and is parallel to y-axis. Let for $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, the shortest distance between the two lines be 3 . Then the square of the distance of the point $(\lambda_1, \lambda_2, 7)$ from the line L_1 is

(A) 25 (B) 32 (C) 40 (D) 37

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Q39. MCQ (Single Correct Answer)

Let g be a differentiable function such that $\int_0^x g(t)dt = x - \int_0^x tg(t)dt, x \geq 0$ and let $y = y(x)$ satisfy the differential equation $\frac{dy}{dx} - y\tan x = 2(x+1)\sec x g(x), x \in \left[0, \frac{\pi}{2}\right)$. If $y(0) = 0$, then $y\left(\frac{\pi}{3}\right)$ is equal to

(A) $\frac{4\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{2\pi}{3\sqrt{3}}$ (D) $\frac{4\pi}{3\sqrt{3}}$

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Q40. MCQ (Single Correct Answer)

The sum of all rational terms in the expansion of $(2+\sqrt{3})^8$ is :

(A) 16923 (B) 18817 (C) 3763 (D) 33845

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Q41. MCQ (Single Correct Answer)

If $\sum_{r=1}^{9}\left(\frac{r+3}{2^r}\right)\cdot {}^9C_r = \alpha\left(\frac{3}{2}\right)^9 - \beta, \alpha, \beta \in \mathbb{N}$, then $(\alpha+\beta)^2$ is equal to

(A) 27 (B) 81 (C) 18 (D) 9

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Q42. MCQ (Single Correct Answer)

Let $z \in C$ be such that $\frac{z^2+3i}{z-2+i} = 2+3i$. Then the sum of all possible values of z^2 is :

(A) $-19+2i$ (B) $-19-2i$ (C) $19-2i$ (D) $19+2i$

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Q43. MCQ (Single Correct Answer)

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}, x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to

(A) 28 (B) 27 (C) -1 (D) 1

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Q44. MCQ (Single Correct Answer)

A line passes through the origin and makes equal angles with the positive coordinate axes. It intersects the lines $\mathrm{L}_1 : 2x + y + 6 = 0$ and $\mathrm{L}_2 : 4x + 2y - p = 0, p > 0$, at the points A and B , respectively. If $AB = \frac{9}{\sqrt{2}}$ and the foot of the perpendicular from the point A on the line L_2 is M, then $\frac{AM}{BM}$ is equal to

(A) 5 (B) 3 (C) 2 (D) 4

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Q45. MCQ (Single Correct Answer)

Let A be a matrix of order 3×3 and $|A| = 5$. If $|2\,\mathrm{adj}(3A\,\mathrm{adj}(2A))| = 2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma}, \alpha, \beta, \gamma \in N$, then $\alpha + \beta + \gamma$ is equal to

(A) 26 (B) 27 (C) 25 (D) 28

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Q46. Numerical

All five letter words are made using all the letters A, B, C, D, E and arranged as in an English dictionary with serial numbers. Let the word at serial number n be denoted by $\mathrm{W_n}$. Let the probability $\mathrm{P}\,(\mathrm{W_n})$ of choosing the word $\mathrm{W_n}$ satisfy $\mathrm{P}\,(\mathrm{W_n}) = 2\mathrm{P}\,(\mathrm{W_{n-1}}), \mathrm{n} > 1$.

If $\mathrm{P(CDBEA)} = \frac{2^{\alpha}}{2^{\beta}-1}, \alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to :_____

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Q47. Numerical

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}, \vec{c} = \lambda\hat{j} + \mu\hat{k}$ and $\hat{d}$ be a unit vector such that $\vec{a} \times \hat{d} = \vec{b} \times \hat{d}$ and $\vec{c} \cdot \hat{d} = 1$. If $\vec{c}$ is perpendicular to $\vec{a}$, then $|3\lambda\hat{d} + \mu\vec{c}|^2$ is equal to_____

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Q48. Numerical

Let the product of the focal distances of the point $\mathbf{P}(4, 2\sqrt{3})$ on the hyperbola $\mathrm{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be 32 . Let the length of the conjugate axis of H be p and the length of its latus rectum be q. Then $p^2 + q^2$ is equal to_____

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Q49. Numerical

The area of the region bounded by the curve $y = \max\{|x|, x|x-2|\}$, the x-axis and the lines $x = -2$ and $x = 4$ is equal to_____

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Q50. Numerical

If the number of seven-digit numbers, such that the sum of their digits is even, is $m \cdot n \cdot 10^n; m, n \in \{1, 2, 3, \ldots, 9\}$, then $m + n$ is equal to_____

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

A gas is kept in a container having walls which are thermally non-conducting. Initially the gas has a volume of $800\ \mathrm{cm}^3$ and temperature $27^\circ\mathrm{C}$. The change in temperature when the gas is adiabatically compressed to $200\ \mathrm{cm}^3$ is:

(Take $\gamma = 1.5$; γ is the ratio of specific heats at constant pressure and at constant volume)

(A) 300 K (B) 600 K (C) 327 K (D) 522 K

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Q52. MCQ (Single Correct Answer)

A person measures mass of 3 different particles as $435.42\ \mathrm{g}, 226.3\ \mathrm{g}$ and 0.125 g . According to the rules for arithmetic operations with significant figures, the addition of the masses of 3 particles will be.

(A) 661.845 g (B) 661.84 g (C) 662 g (D) 661.8 g

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Q53. MCQ (Single Correct Answer)

A wire of length 25 m and cross-sectional area $5\ \mathrm{mm}^2$ having resistivity of $2 \times 10^{-6}\Omega\ \mathrm{m}$ is bent into a complete circle. The resistance between diametrically opposite points will be

(A) 100Ω (B) 2.5Ω (C) 12.5Ω (D) 50Ω

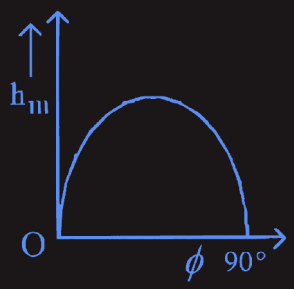
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Q54. MCQ (Single Correct Answer)

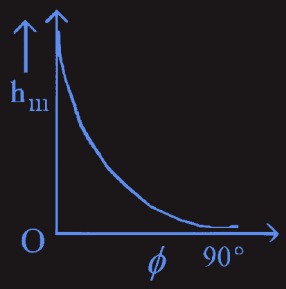
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The angle of projection of a particle is measured from the vertical axis as ϕ and the maximum height reached by the particle is h_m. Here h_m as function of ϕ can be presented as

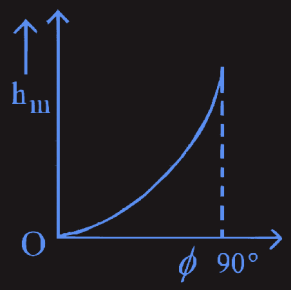
(A)



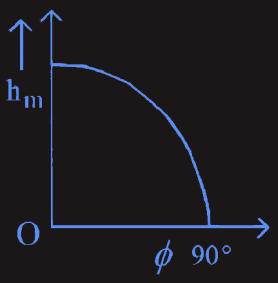
(B)



(C)



(D)



Q55. MCQ (Single Correct Answer)

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The work function of a metal is 3 eV . The color of the visible light that is required to cause emission of photoelectrons is

(A) Red (B) Green (C) Blue (D) Yellow

Q56. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List - I		List - II	
A.	Gravitational constant	I.	$\left[\mathrm{LT}^{-2}\right]$
B.	Gravitational potential energy	II.	$\left[\mathrm{L}^{2}\,\mathrm{T}^{-2}\right]$
C.	Gravitational potential	III.	$\left[\mathrm{ML}^{2}\,\mathrm{T}^{-2}\right]$
D.	Acceleration due to gravity	IV.	$\left[\mathrm{M}^{-1}\,\mathrm{L}^{3}\,\mathrm{T}^{-2}\right]$

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I

(B) A-II, B-IV, C-III, D-I

(C) A-I, B-III, C-IV, D-II

(D) A-III, B-II, C-I, D-IV

Q57. MCQ (Single Correct Answer)

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Consider a completely full cylindrical water tank of height 1.6 m and of cross-sectional area $0.5\ \mathrm{m}^2$. It has a small hole in its side at a height 90 cm from the bottom. Assume, the cross-sectional area of the hole to be negligibly small as compared to that of the water tank. If a load 50 kg is applied at the top surface of the water in the tank then the velocity of the water coming out at the instant when the hole is opened is:

$\left(\mathrm{g} = 10\ \mathrm{m/s}^2\right)$

(A) $2\ \mathrm{m/s}$

(B) $5\ \mathrm{m/s}$

(C) $3\ \mathrm{m/s}$

(D) $4\ \mathrm{m/s}$

Q58. MCQ (Single Correct Answer)

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Consider following statements for refraction of light through prism, when angle of deviation is minimum.

A. The refracted ray inside prism becomes parallel to the base.

B. Larger angle prisms provide smaller angle of minimum deviation.

C. Angle of incidence and angle of emergence becomes equal.

D. There are always two sets of angle of incidence for which deviation will be same except at minimum deviation setting.

E. Angle of refraction becomes double of prism angle.

Choose the correct answer from the options given below :

(A) A, B and E Only

(B) B, C and D Only

(C) B, D and E Only

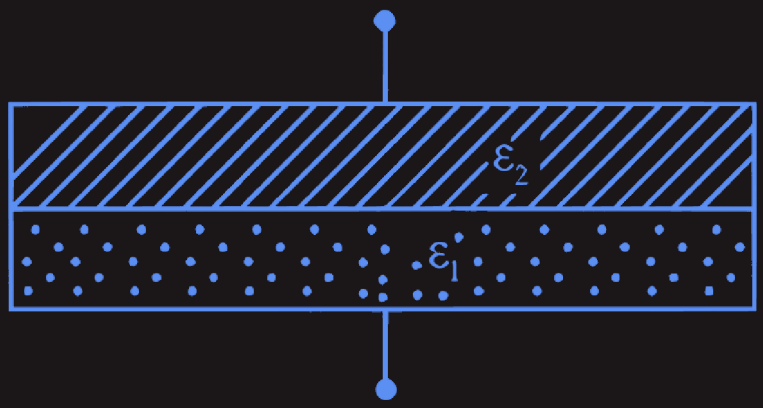
(D) A, C and D Only

Q59. MCQ (Single Correct Answer)

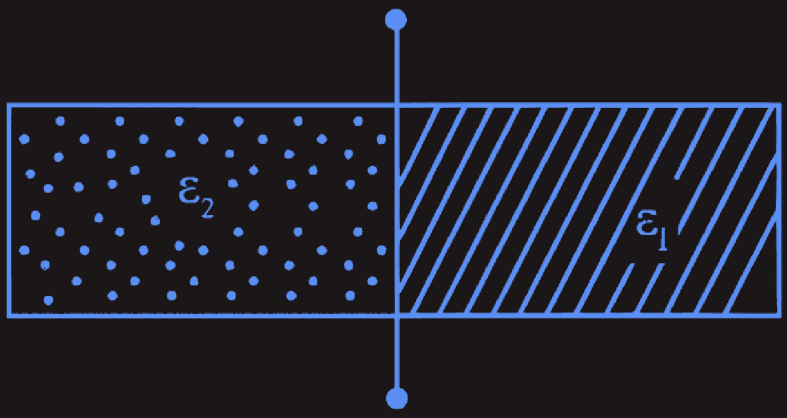
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A parallel plate capacitor is filled equally(half) with two dielectrics of dielectric constants ε_1 and ε_2, as shown in figures. The distance between the plates is d and area of each plate is A. If capacitance in first configuration and second configuration are C_1 and C_2 respectively, then $\frac{C_1}{C_2}$ is:

First Configuration



Second Configuration



(A) $\frac{\varepsilon_0(\varepsilon_1+\varepsilon_2)}{2}$

(B) $\frac{\varepsilon_1\varepsilon_2^2}{(\varepsilon_1+\varepsilon_2)^2}$

(C) $\frac{4\varepsilon_1\varepsilon_2}{(\varepsilon_1+\varepsilon_2)^2}$

(D) $\frac{\varepsilon_1\varepsilon_2}{\varepsilon_1+\varepsilon_2}$

Q60. MCQ (Single Correct Answer)

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During the melting of a slab of ice at 273 K at atmospheric pressure :

(A) Internal energy of ice-water system remains unchanged.

(B) Positive work is done by the ice-water system on the atmosphere.

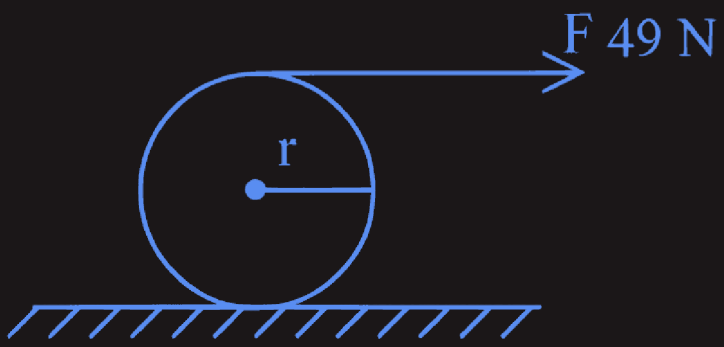
(C) Positive work is done on the ice-water system by the atmosphere.

(D) Internal energy of the ice-water system decreases.

Q61. MCQ (Single Correct Answer)

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A force of 49 N acts tangentially at the highest point of a sphere (solid) of mass 20 kg , kept on a rough horizontal plane. If the sphere rolls without slipping, then the acceleration of the center of the sphere is



(A) $2.5\ \mathrm{m/s^2}$

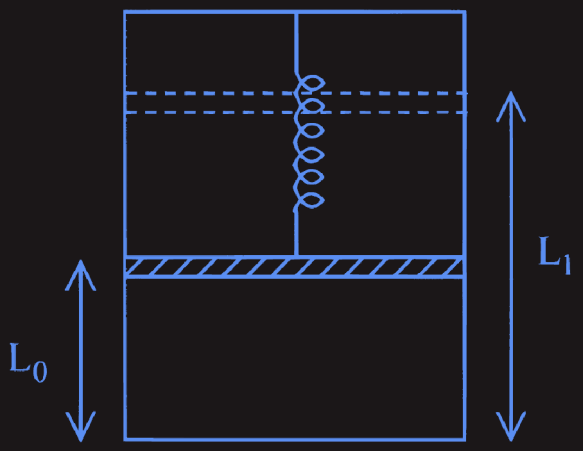
(B) $3.5\ \mathrm{m/s^2}$

(C) $0.25\ \mathrm{m/s^2}$

(D) $0.35\ \mathrm{m/s^2}$

Q62. MCQ (Single Correct Answer)

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A piston of mass M is hung from a massless spring whose restoring force law goes as $F=-kx^3$, where k is the spring constant of appropriate dimension. The piston separates the vertical chamber into two parts, where the bottom part is filled with ' n ' moles of an ideal gas. An external work is done on the gas isothermally (at a constant temperature T) with the help of a heating filament (with negligible volume) mounted in lower part of the chamber, so that the piston goes up from a height $\mathrm{L_0}$ to $\mathrm{L_1}$, the total energy delivered by the filament is:(Assume spring to be in its natural length before heating)

(A) $nRT\ln\left(\frac{L_1}{L_0}\right)+Mg\left(L_1-L_0\right)+\frac{3k}{4}\left({L_1}^4-{L_0}^4\right)$

(B) $nRT\ln\left(\frac{L_1}{L_0}\right)+Mg\left(L_1-L_0\right)+\frac{k}{4}\left(L_1^4-{L_0}^4\right)$

(C) $nRT\ln\left(\frac{L_1^2}{L_0^2}\right)+\frac{Mg}{2}\left(L_1-L_0\right)+\frac{k}{4}\left(L_1^4-{L_0}^4\right)$

(D) $3nRT\ln\left(\frac{L_1}{L_0}\right)+2Mg\left(L_1-L_0\right)+\frac{k}{3}\left({L_1}^3-{L_0}^3\right)$

Q63. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List - I		List - II	
A.	${}_0^1\mathrm{n} + {}_{92}^{235}\mathrm{U} \rightarrow {}_{54}^{140}\mathrm{Xe} + {}_{38}^{94}\mathrm{Sr} + 2{}_0^1\mathrm{n}$	I.	Chemical reaction
B.	$2\mathrm{H}_2 + \mathrm{O}_2 \rightarrow 2\mathrm{H}_2\mathrm{O}$	II.	Fusion with +ve Q value
C.	${}_1^2\mathrm{H} + {}_1^2\mathrm{H} \rightarrow {}_2^3\mathrm{He} + {}_0^1\mathrm{n}$	III.	Fission
D.	${}_1^1\mathrm{H} + {}_1^3\mathrm{H} \rightarrow {}_1^2\mathrm{H} + {}_1^2\mathrm{H}$	IV.	Fusion with -ve Q value

Choose the correct answer from the options given below:

(A) A-II, B-I, C-III, D-IV

(B) A-III, B-I, C-II, D-IV

(C) A-III, B-I, C-IV, D-II

(D) A-II, B-I, C-IV, D-III

Q64. MCQ (Single Correct Answer)

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A particle is released from height S above the surface of the earth. At certain height its kinetic energy is three times its potential energy. The height from the surface of the earth and the speed of the particle at that instant are respectively.

(A) $\frac{S}{4}, \frac{3gS}{2}$

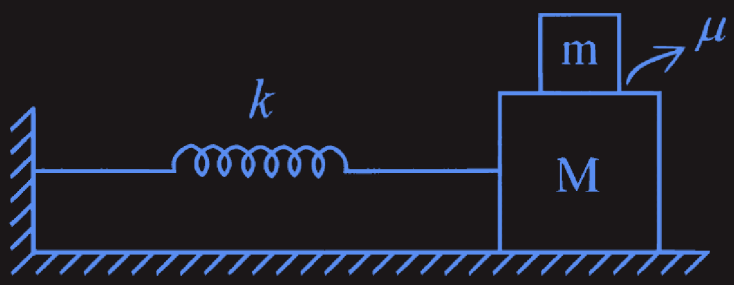
(B) $\frac{S}{2}, \frac{3gS}{2}$

(C) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

(D) $\frac{S}{2}, \sqrt{\frac{3gS}{2}}$

Q65. MCQ (Single Correct Answer)

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Two blocks of masses m and $M, (M > m)$, are placed on a frictionless table as shown in figure. A massless spring with spring constant k is attached with the lower block. If the system is slightly displaced and released, then (μ = coefficient of friction between the two blocks)

A. The time period of small oscillation of the two blocks is $T = 2\pi\sqrt{\frac{(m+M)}{k}}$

B. The acceleration of the blocks is $a = -\frac{kx}{M+m}$ (x = displacement of the blocks from the mean position)

C. The magnitude of the frictional force on the upper block is $\frac{m\mu|x|}{M+m}$

D. The maximum amplitude of the upper block, if it does not slip, is $\frac{\mu(M+m)g}{k}$

E. Maximum frictional force can be $\mu(\mathrm{M}+\mathrm{m})\mathrm{g}$.

Choose the correct answer from the options given below :

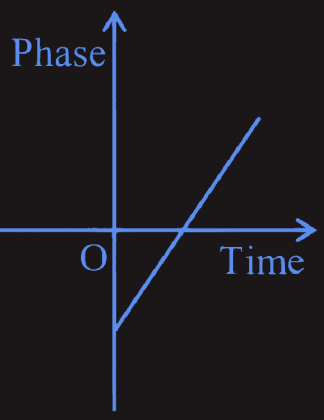
(A) B, C, D Only (B) C, D, E Only (C) A, B, D Only (D) A, B, C Only

Q66. MCQ (Single Correct Answer)

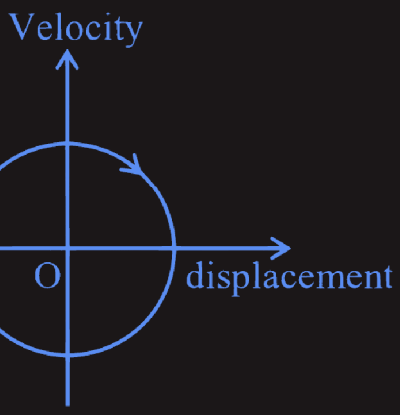
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Which of the following curves possibly represent one-dimensional motion of a particle?

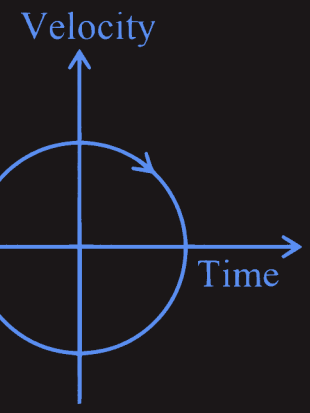
(A)

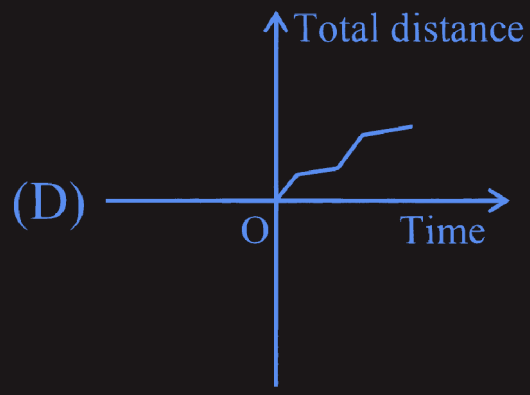


(B)



(C)





Choose the correct answer from the options given below :

(A) A, B and C only (B) A, C and D only (C) A, B and D only (D) A and B only

Q67. MCQ (Single Correct Answer)

The radii of curvature for a thin convex lens are 10 cm and 15 cm respectively. The focal length of the lens is 12 cm . The refractive index of the lens material is

(A) 1.4 (B) 1.8 (C) 1.5 (D) 1.2

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Q68. MCQ (Single Correct Answer)

The electrostatic potential on the surface of uniformly charged spherical shell of radius $\mathrm{R} = 10\ \mathrm{cm}$ is 120 V . The potential at the centre of shell, at a distance $\mathrm{r} = 5\ \mathrm{cm}$ from centre, and at a distance $\mathrm{r} = 15$ cm from the centre of the shell respectively, are:

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(A) $0\ \mathrm{V}, 120\ \mathrm{V}, 40\ \mathrm{V}$ (B) $120\ \mathrm{V}, 120\ \mathrm{V}, 80\ \mathrm{V}$

(C) $40\ \mathrm{V}, 40\ \mathrm{V}, 80\ \mathrm{V}$ (D) $0\ \mathrm{V}, 0\ \mathrm{V}, 80\ \mathrm{V}$

Q69. MCQ (Single Correct Answer)

The radiation pressure exerted by a 450 W light source on a perfectly reflecting surface placed at 2 m away from it, is

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(A) 3×10^{-8} Pascals (B) 0

(C) 1.5×10^{-8} Pascals (D) 6×10^{-8} Pascals

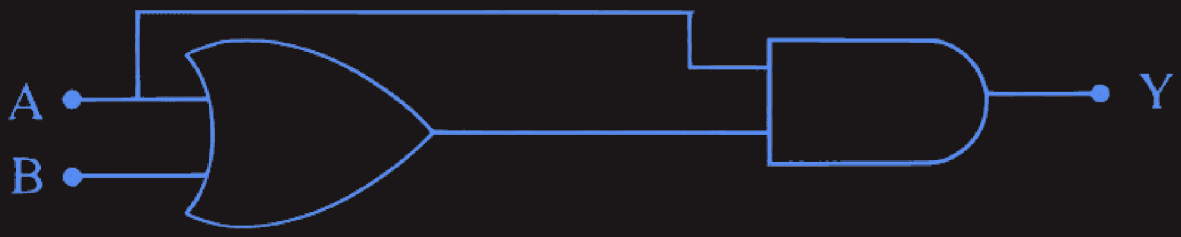
Q70. MCQ (Single Correct Answer)

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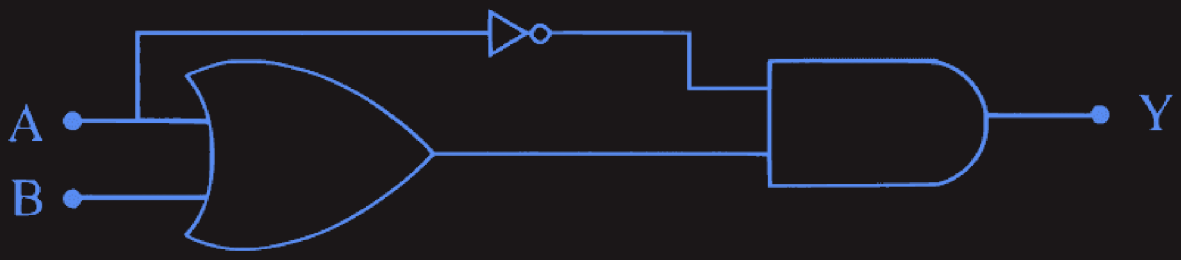
Choose the correct logic circuit for the given truth table having inputs A and B.

Inputs		Output
A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

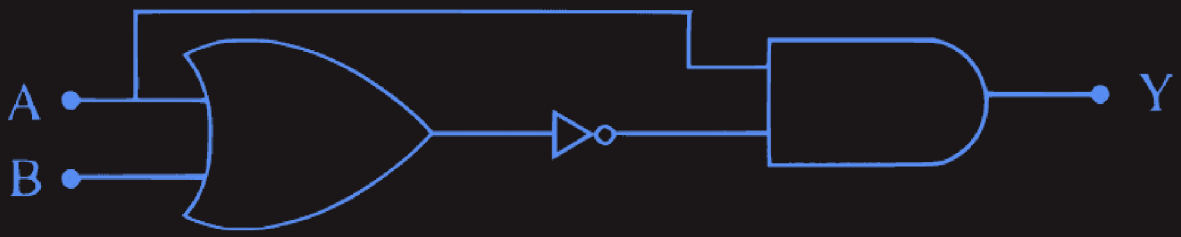
(A)



(B)



(C)



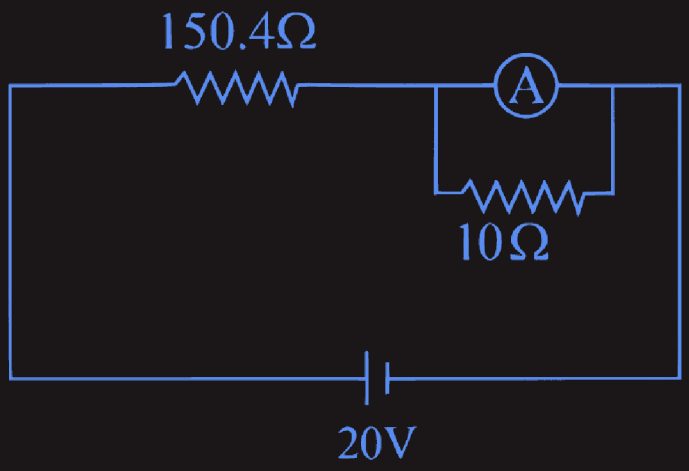
(D)



Q71. Numerical

In the figure shown below, a resistance of 150.4Ω is connected in series to an ammeter A of resistance 240Ω. A shunt resistance of 10Ω is connected in parallel with the ammeter. The reading of the ammeter is_____mA .

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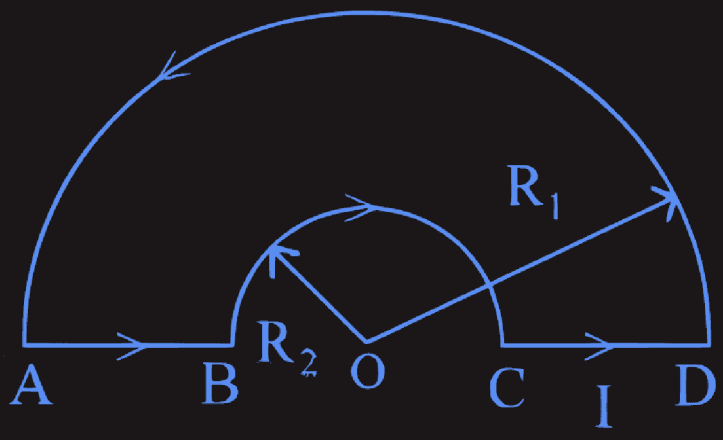


Q72. Numerical

A loop ABCDA , carrying current $\mathrm{I} = 12\ \mathrm{A}$, is placed in a plane, consists of two semi-circular segments of radius $R_1 = 6\pi\ \mathrm{m}$ and $\mathrm{R}_2 = 4\pi\ \mathrm{m}$. The magnitude of the resultant magnetic field at center O is $\mathrm{k} \times 10^{-7}\ \mathrm{T}$. The value of k is_____.

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(Given $\mu_0 = 4\pi \times 10^{-7}\mathrm{TmA}^{-1}$)



Q73. Numerical

Two coherent monochromatic light beams of intensities 4I and 9I are superimposed. The difference between the maximum and minimum intensities in the resulting interference pattern is $x\mathrm{I}$. The value of x is_____ .

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Q74. Numerical

Three identical spheres of mass m , are placed at the vertices of an equilateral triangle of length a. When released, they interact only through gravitational force and collide after a time $\mathrm{T} = 4$ seconds. If the sides of the triangle are increased to length $2a$ and also the masses of the spheres are made 2 m , then they will collide after_____seconds.

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Q75. Numerical

A 4.0 cm long straight wire carrying a current of 8 A is placed perpendicular to a uniform magnetic field of strength 0.15 T . The magnetic force on the wire is _____mN .

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Answer Key

Chemistry

1. **C**	2. **B**	3. **C**	4. **C**	5. **A**
6. **C**	7. **C**	8. **C**	9. **A**	10. **A**
11. **C**	12. **D**	13. **B**	14. **A**	15. **B**
16. **B**	17. **A**	18. **B**	19. **B**	20. **A**
21. **109**	22. **6**	23. **80**	24. **2**	25. **25**

Mathematics

26. **A**	27. **B**	28. **A**	29. **B**	30. **A**
31. **B**	32. **A**	33. **A**	34. **C**	35. **C**
36. **B**	37. **C**	38. **A**	39. **A**	40. **B**
41. **B**	42. **B**	43. **C**	44. **B**	45. **B**
46. **183**	47. **5**	48. **120**	49. **12**	50. **14**

Physics

51. **A**	52. **D**	53. **B**	54. **D**	55. **C**
56. **A**	57. **D**	58. **D**	59. **C**	60. **C**
61. **B**	62. **B**	63. **B**	64. **C**	65. **C**
66. **C**	67. **C**	68. **B**	69. **D**	70. **A**
71. **5**	72. **1**	73. **24**	74. **8**	75. **48**

For numerical answers given as a range, any value within the range is correct.

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